CUNY AI LAB

# Configuring skills and tools

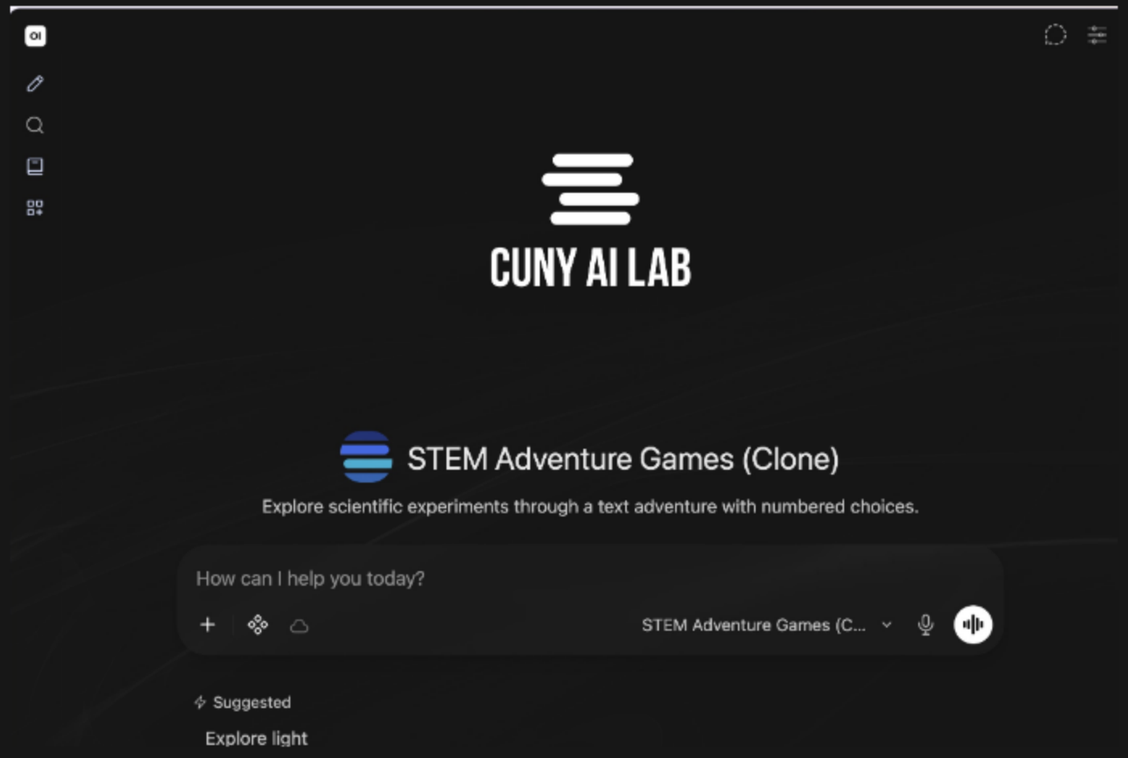
New Media Lab · Room 7388.01
CUNY Graduate Center

Thursday, October 15, 2026 · 2:30–4:00 p.m. Eastern

https://cuny-ai-lab.github.io/sandbox-series/skills

# Choose Your Own Adventure

Continue your saved model, or clone **STEM Adventure Games**
Read its system prompt for setting, goals, and rules
Clone instructions · Adventure example



# What Is a Skill?

**Skills** contain plain-text instructions for repeatable tasks

| | |
|---|---|
| **Trigger** | Request or condition that starts a procedure |
| **Procedure** | Steps, checks, and points that require a question |
| **Output** | Required response content and form |

# Skill Examples

Workspace → Skills · 1 of 3

Examples from the host's catalog. Your list depends on access

**History Of Ideas Abstract Critique** /history-of-ideas-abstract-critique 3 days ago

Guide revision of relationships among ideas and their implications.

**Historical Context Abstract Critique** /historical-context-abstract-critique 3 days ago

Guide revision of historical conditions as drivers of intellectual change.

**Extend STEM Adventures** /extend-stem-adventures a month ago

Add a source-based experimental procedure to STEM Adventure, or examine a submitted play record.

**ChartJS Visualization** /chartjs-visualization 2 months ago

Use whenever a request would benefit from plotting quantitative data or mathematical relationships, even when the user ...

# Skill Examples

Workspace → Skills · 2 of 3

Examples from the host's catalog. Your list depends on access

Response Features /response-features 5 months ago

Error Handling /error-handling 5 months ago

Language Adjustment /language-adjustment- 5 months ago

Writing-Support-Opener /writing-support-opener 6 months ago

Open every interaction with one brief, humanizing question unrelated to the assignment

# Skill Examples

Workspace → Skills · 3 of 3

Examples from the host's catalog. Your list depends on access

**Conversational Etiquette** /conversational-etiquette 6 months ago

Supports short, turn-taking dialogue as a prose-first, anti-slop corrective

**Native Orthography** /preserve-native-orthography 6 months ago

Preserves correct native spelling, scripts, diacritics, and punctuation across languages.

**One-Step Dialogue** /adaptive_conversational_scaffolding_v1 6 months ago

Pedagogical scaffolding for inquisitive dialogue following a one-step extension + question per turn

# Add Your Skill

Open **Integrations → Skills** in your model's chat

Find a saved skill for your task and turn on its toggle
Read its trigger, procedure, and expected response

Selection guide · Current workspace examples

## Add Skill Instructions

Add this clause with the saved skill's task, name, and ID · Copy clause

When I ask for [TASK], load [SKILL_NAME] (ID: [SKILL_ID]). Follow its saved instructions using this system prompt's goals and rules and the latest information I have supplied. Ask about missing or conflicting details before deciding. Acknowledge a request to stop immediately, without requiring another action or reflection.

Save your model, then check that the saved skill loads and follows its instructions

# Test Your Skill

Ask your model to use the selected skill for its stated task

Use an actual record of choices and results
Leave out required information and check for a question
Ask to stop before the goal is met

Which task in your teaching or research needs a repeatable procedure?

# What Is a Tool?

When a model calls a **tool**, Sandbox executes a function and returns its result

**Tool calling**, otherwise known as **function calling**, sends a function name and arguments for execution

| | |
|---|---|
| **Input** | Arguments supplied to a function |
| **Execution** | Code that applies specified rules |
| **Result** | Returned output for a model to inspect |

# Tool Examples

Workspace → Tools · 1 of 4

Examples from the host's catalog. Your list depends on access

**Adventure Kit** v2.0.0 8 hours ago

Create participant-defined choice adventures with configurable state, rules and endings. No built-in scenario.

**Chat Export** v0.3.1 21 hours ago

Saves chat content (headings, text, lists and tables) as a downloadable Word (.docx) or PDF file, plus Markdow…

**Instructor Rubric** v0.4.0 21 hours ago

Checks student lab work without revealing answers. Verifies ONE calculation per lab part, flags data that break…

# Tool Examples

Workspace → Tools · 2 of 4

Examples from the host's catalog. Your list depends on access

### OCR reader v0.2.0 2 days ago

reads scanned or image-only PDFs (and images) that come up blank. Runs OCR on pages with no text layer and...

### Instructional Toolset v0.6.0 3 days ago

Deterministically lists and loads curriculum units for instructional AI models.

### Pronunciation and Translation Tool ✨ v5.1.1 3 days ago

https://openwebui.com/posts/pronunciation_and_translation_tool_d8a83bca

# Tool Examples

Workspace → Tools · 3 of 4

Examples from the host's catalog. Your list depends on access

**Graphing** v0.1.0 a month ago

Graphing tool

**PhysicsPY** v0.4.0 2 months ago

Source-grounded physics laboratory analysis with static SVG charts.

**ChartJS** v2.0.0 2 months ago

Charts is a tool for the Sandbox that renders chart elements in-chat using an HTML5 canvas via Chart.js cdn

**Charts (v1)** v1.3.0 2 months ago

Charts is a tool for OpenWebUI that allows you to create charts using the Chart.js cdn

# Tool Examples

Workspace → Tools · 4 of 4

Examples from the host's catalog. Your list depends on access

**Preflight Student Input** 3 months ago

Solicits input from students before commencing an exercise or assignment

**Kale GitHub** v0.1.0 5 months ago

Checks recent activity across CUNY AI Lab's GitHub org

**Whisper CUDA** v0.7.0 a year ago

Whisper Cuda

**cloudflare_whisper_tool** v0.6.0 a year ago

cloudflare_whisper_tool

# Check Tool Binding

**Model Bound Tool Enabler** is a filter that restores model-attached tools when a chat request omits them

If an attached tool is missing from call details, use this existing filter on your private model and repeat your test

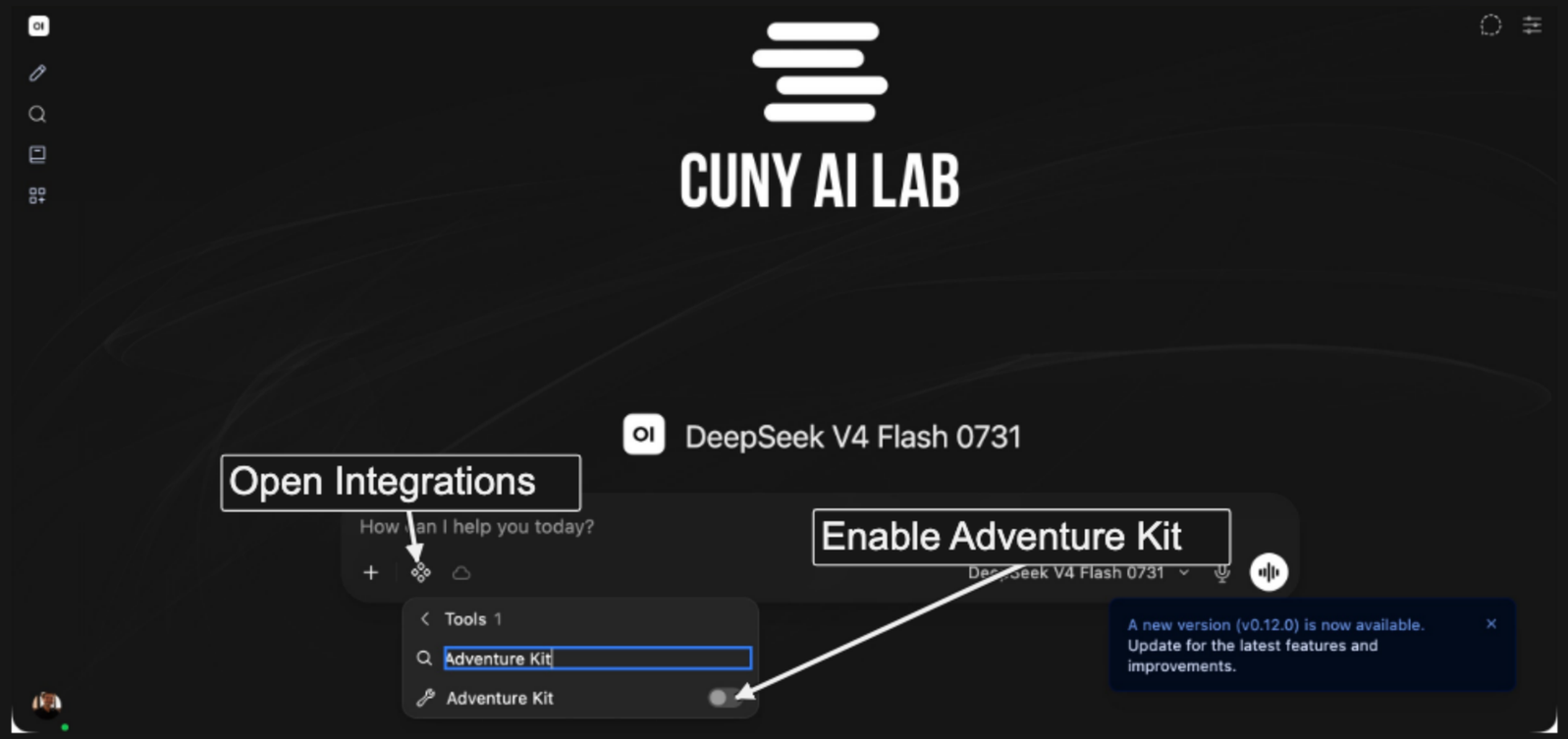
It restores availability; execution still requires a valid tool call

Configuration checks and evidence record

# Add the ‘Adventure Kit’ tool

Open **Integrations → Tools**

Search **Adventure Kit** and select its toggle



# Prepare Your Adventure

Design your adventure from your system prompt

Instruct your model to map its setting, goals, and rules into scenes, choices, and end conditions

Resolve missing or contradictory instructions
Review proposed input for **Adventure Kit**

Copy preparation and run requests

# Add Tool Instructions

Add this clause to your model's system prompt · Copy clause

After the player approves an adventure, call the attached Adventure Kit tool's **render_adventure** function with the reviewed schema-version-2 JSON in **scenario_json**. Derive content and rules from this system prompt; resolve missing or conflicting instructions first. Read the returned result, resolve errors before play, and use the player's copied Play record when discussing progress or completion.

Save your model, then verify tool attachment, provider support, and the actual result

## Test Your Tool

Run your reviewed adventure, then compare its scene and choices with your prompt

Try a choice before and after its prerequisite
Inspect Undo and Stop
Copy actual play record into chat

Which step in your teaching or research needs an executable operation?

# Choose Your Task

## Revise One Item

Copy complete tool code or skill instructions

Describe an observed problem and one change

Copy tool code · Skill instructions

## Create Your Own

Choose your own tool or skill

Specify purpose, required input, and expected result

# Work With Support

Open **CAIL Workspace Support**
Paste original content or your new task, system prompt, and required sources
Request complete content, an explanation, and one test · Copy request



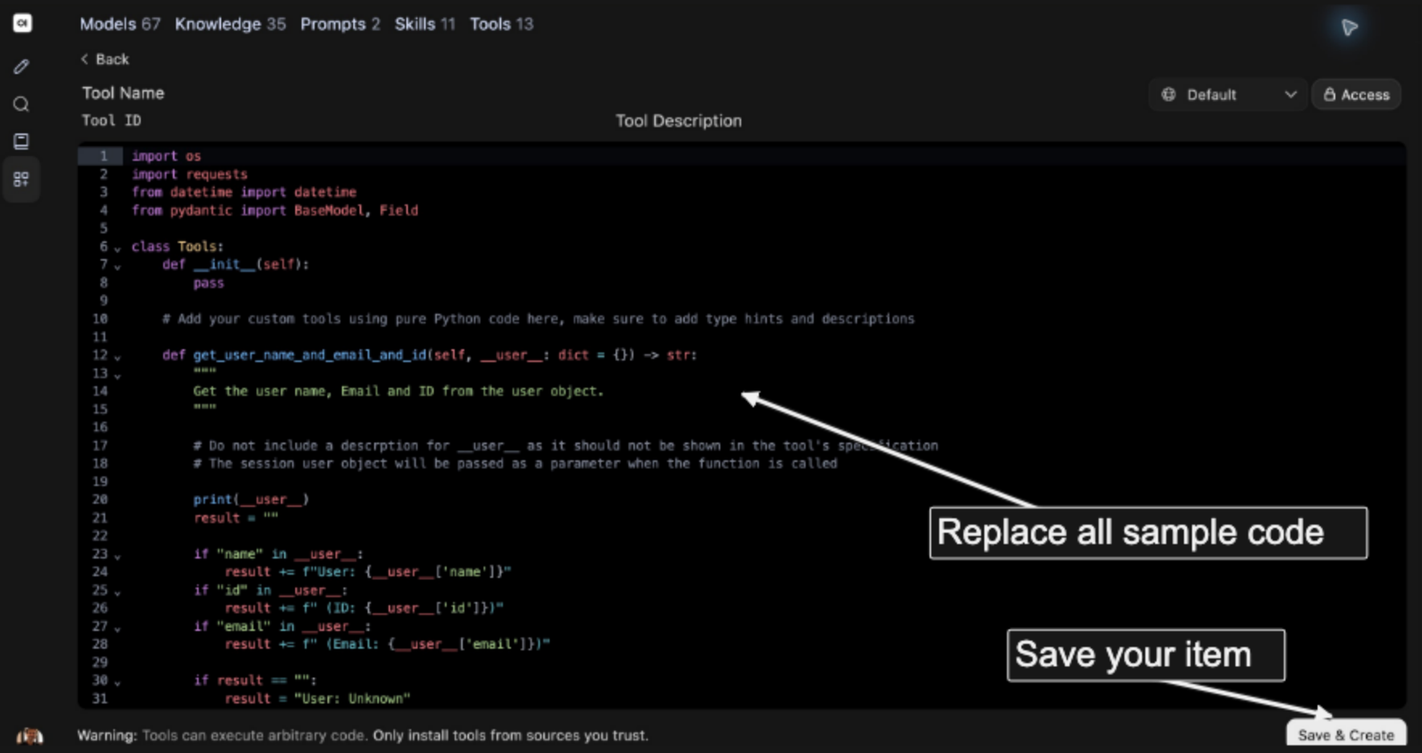
# Add Tools or Skills

Create a new **tool** or **skill**, or save a revised copy

Use **Workspace → Tools → Create** or **Skills → Create**

Enter new Name and ID; paste complete reviewed content

Review **Access**, then **Save & Create** · Select and test your item



## Series Reflections

What task in your teaching or research could benefit from a skill or tool?

Would you be interested in joining our coworking session on October 29, 2–4pm, to build your adventure into a collection so that we can showcase our work over the course of the series?

What would you develop next with the prompts, knowledge collections, skills, and tools you have explored?